ENVS193DS Final

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GitHub repo

Access my repo here.

Set up

```
# read in packages
library(tidyverse) # general use
library(here) # file organization
library(janitor) # cleaning data frames
library(DHARMa) # for GLM diagnostics
library(ggeffects) # for ggpredict
library(flextable) # for flextable
library(gtsummary) # for GLM
library(MuMIn) # for AIC

# read in data
nests <- read_csv(here("data", "nest_data_final.csv"))
# read in personal data to object from csv
my_data <- read_csv(here("data", "envs193ds-personal-data.csv"))
```

```
# setting up personal data for problem 3

# cleaned column names
my_data <- my_data |>
  clean_names()

# created new column: 1 if workout was a run, 0 if not
my_data <- my_data |>
  mutate(is_run = case_when(
```

```
    workout_type == "Run" ~ 1, # run = 1
    .default = 0))             # all other workout types = 0
```

Problem 1. Research writing

a. Transparent statistical methods

Part 1: The statistical test used in Part 1 was a generalized linear model (logistic regression) with a binomial distribution. The response variable is a binary outcome to reflect the probability that the American Avocet uses the mircohabitat (1) or doesn't (0). The predictor is a continuous variable: distance to water edge (UNITS).

Part 2: The statistical test used in Part 2 was a chi-square test of independence which is used to determine whether there is a relationship between two categorical variables. The categorical variables are bird species (American Avocet, Forster's Tern, Black-necked Stilt) and habitat type (managed pond, non-tidal marsh, salt pond, tidal marsh, tidal mudflat). The test compares observed counts to expected counts.

b. Suggestions for rewriting

Part 1: American Avocets were more likely to use microhabitats that were closer to the water's edge, and this relationship was strong enough that is unlikely to be the result of random chance alone.

Part 2: The three shorebird species did not use the five habitat types in equal proportions: each species showed a distinct pattern of habitat preference, and these differences across species are unlikely to reflect chance.

c. Further analysis

Tidal inundation frequency could influence American Avocet microhabitat use because Avocets forage in shallow flooded areas so microhabitats that are too rarely or too frequently inundated may not provided the water depth conditions needed for successful foraging. Tidal inundation frequency is a continuous variable, measured as the proportion of time a microhabitat is flooded per tidal cycle; this variable could be recorded using tidal monitoring stations.

Problem 2. Data analysis

a. Clean and wrangle

```
# created new object from nests
nests_clean <- nests |>
# filtered to include ant species: C. mimosae, C. sjostedti, and C. nigriceps
filter(ant_species %in% c("RRB", "AB", "BBR")) |>
# created new column with scientific name
mutate(species_name = recode_values(ant_species,
                                    "RRB" ~ "Crematogaster mimosae",
                                    "AB" ~ "Crematogaster sjostedti",
                                    "BBR" ~ "Crematogaster nigriceps")) |>
# reordered ant species
mutate(species_name = as_factor(species_name),
       species_name = fct_relevel(species_name,
                                  "Crematogaster sjostedti",
                                  "Crematogaster mimosae",
                                  "Crematogaster nigriceps")) |>
# selected columns of interest
select(height_cm, case_control, species_name)

# checked structure
str(nests_clean)
```

```
tibble [270 x 3] (S3: tbl_df/tbl/data.frame)
 $ height_cm   : num [1:270] 392 310 513 154 324 ...
 $ case_control: num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ species_name: Factor w/ 3 levels "Crematogaster sjostedti",..: 2 2 2 2 1 2 3 2 1 3 ...
```

```
# displayed ten random rows
slice_sample(nests_clean, # cleaned data frame to display
             n = 10)      # showed ten rows
```

```
# A tibble: 10 x 3
   height_cm case_control species_name
       <dbl>        <dbl> <fct>
 1      136.            0 Crematogaster mimosae
 2      237             1 Crematogaster nigriceps
 3      208             0 Crematogaster mimosae
```

```
 4       99              0 Crematogaster mimosae
 5      512.             1 Crematogaster mimosae
 6      192              0 Crematogaster mimosae
 7      214.             0 Crematogaster mimosae
 8      112              0 Crematogaster nigriceps
 9      123              0 Crematogaster mimosae
10      149              0 Crematogaster mimosae
```

b. Response variable

The response variable `case_control` is binary and describes whether a tree in the survey contained a nest (1) or was an "available" tree for birds to nest in (0).

c. Purpose of study

- Tree height is a continuous predictor variable (centimeters). FINISH WITH CITATION
- Acacia ant species is a categorical predictor variable with three levels: C. sjostedti, C. mimosae, and C. nigriceps. FINISH WITH CITATION

d. Table of models

Model number	Tree Height (cm)	Ant Species	Model description
1	X	X	all predictors, no interaction
2	X	X	all predictors, interaction term

e. Run the models

```
# model 1: no interaction
# formula: case_control as a function of tree height plus ant species
model1 <- glm(case_control ~ height_cm + species_name,
            # data: nests_clean data frame
            data = nests_clean,
            # binomial error distribution for binary response
            family = binomial)

# model 2: interaction between tree height and ant species
# formula: case_control as a function of tree height times ant species
```

```r
model2 <- glm(case_control ~ height_cm * species_name,
            # data: nests_clean data frame
            data   = nests_clean,
            family = binomial)
```

f. Select the best model

```r
# calculated AIC
AICc (model1,
      model2) |>
# arranged output in order of AIC
arrange (AICc)
```

```
       df     AICc
model2  6 238.1236
model1  4 258.8940
```

```r
summary(model2)
```

```
Call:
glm(formula = case_control ~ height_cm * species_name, family = binomial,
    data = nests_clean)

Coefficients:
                                                Estimate Std. Error z value
(Intercept)                                    -3.693410   2.517395  -1.467
height_cm                                       0.003895   0.008237   0.473
species_nameCrematogaster mimosae               0.848334   2.554617   0.332
species_nameCrematogaster nigriceps           -12.279738   5.946723  -2.065
height_cm:species_nameCrematogaster mimosae     0.002597   0.008386   0.310
height_cm:species_nameCrematogaster nigriceps   0.084541   0.031961   2.645
                                              Pr(>|z|)
(Intercept)                                    0.14233
height_cm                                      0.63634
species_nameCrematogaster mimosae              0.73983
species_nameCrematogaster nigriceps            0.03893 *
height_cm:species_nameCrematogaster mimosae    0.75679
height_cm:species_nameCrematogaster nigriceps  0.00817 **
```

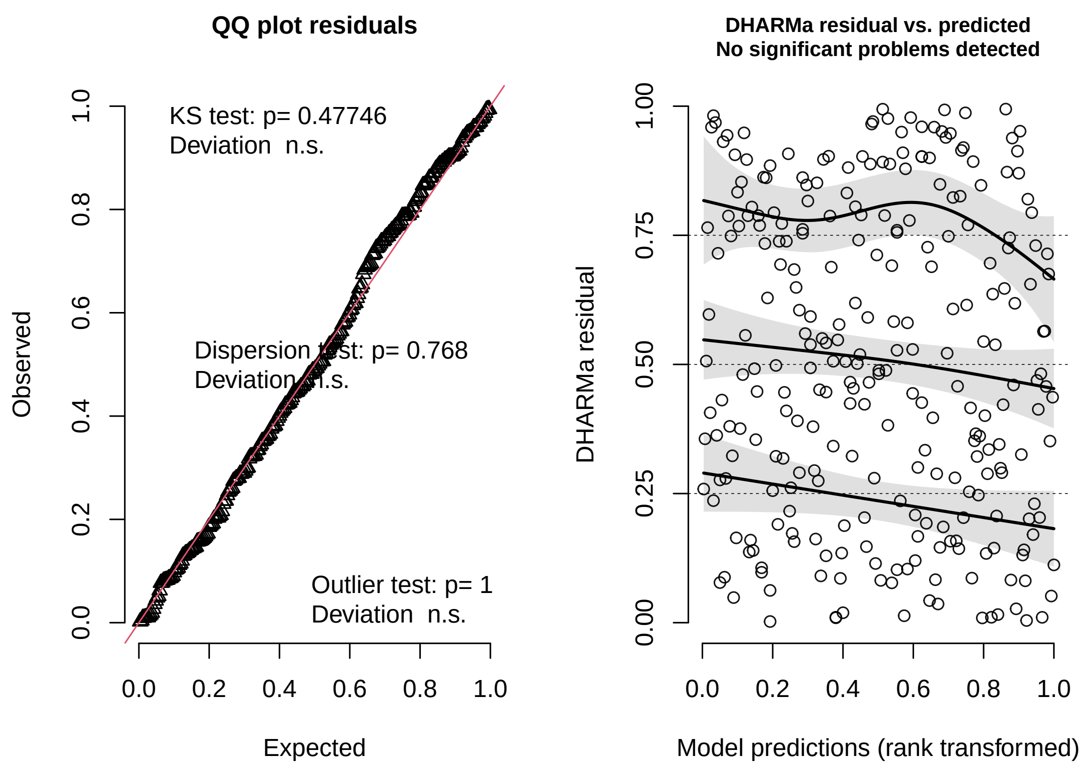
```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 286.04  on 269  degrees of freedom
Residual deviance: 225.80  on 264  degrees of freedom
AIC: 237.8

Number of Fisher Scoring iterations: 7
```

The best model as determined by Akaike's Information Criterion (AIC) is model 2 which shows the presence of nests as a function of tree height multiplied by ant species.

g. Check the diagnostics

```r
# simulated residuals from DHARMa for model 2
plot(simulateResiduals(model2))
```

DHARMa residual



FIX BELOW

Information from left plot (QQ plot of residuals): There appears to be a uniform distribution because the points follow the reference line which matches the quantiles of simulated residuals with quantiles of uniform distribution

- **KS test**: Do my residuals follow a uniform distribution? → Yes
- **Dispersion test**: Are your residuals more dispersed than expected from a uniform distribution? → No
- **Outlier test**: Are there outliers in your data set that could be influencing model coefficients or predictions? → No

Information from right plot (residuals vs predicted values):

- There appear to be randomly arranged residuals across the range of predicted values.

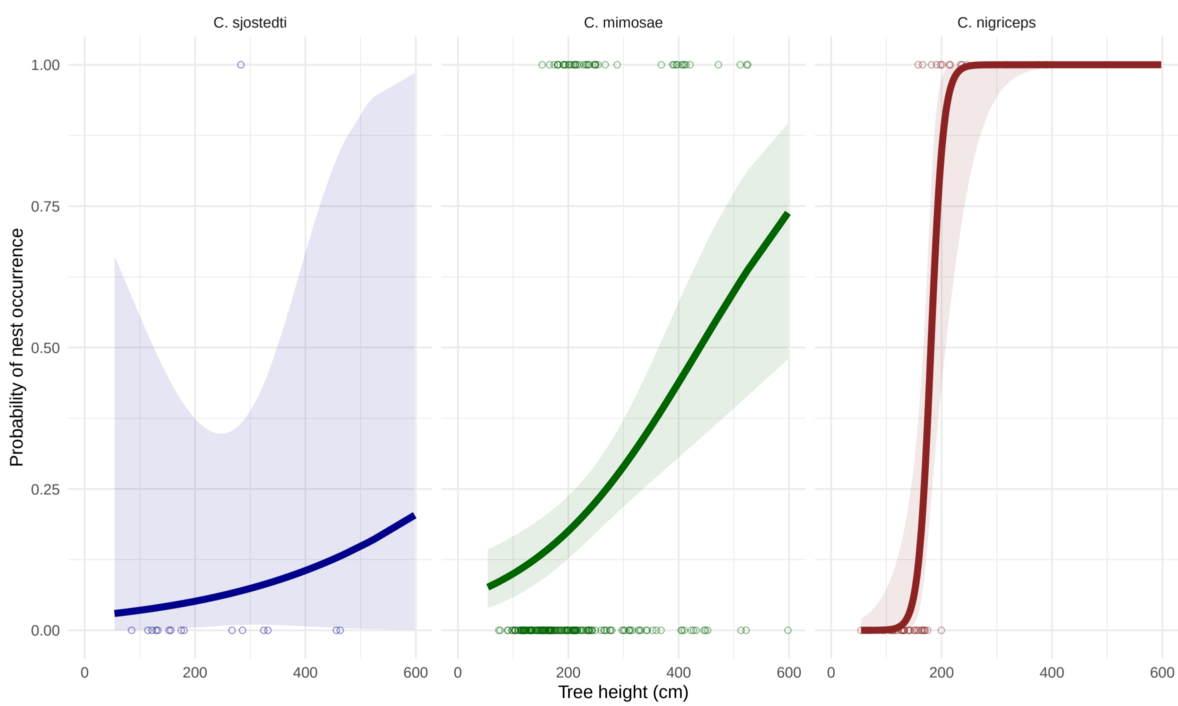
h. Visualize the model predictions

Generated model predictions across range of height values for each ant species

```r
# generated model predictions across range of height values for each ant species
preds <- ggpredict(
  model2, # model object
  terms = c("height_cm [all]", "species_name")) |>  # predictors
  # renamed column to fit original data frame
  rename(height_cm = x,
         species_name = group)
```

```r
# base layer: ggplot
ggplot(data = nests_clean, # data frame: nests_clean
       mapping = aes(x = height_cm, # x-axis: height (cm)
                     y = case_control )) + # y-axis: presense of nest (0/1)
  # first layer: underlying points representing individual observations
  geom_point(aes(color = species_name), # colored points by ant species
             shape = 21, # made the points open circle
             alpha = 0.4) + # made points more transparent
  # second layer: model prediction line
  geom_line(data = preds, # data frame: preds
            mapping = aes(x = height_cm, # x-axis: height (cm)
                          y = predicted, # y-axis: predicted
                          color = species_name), # color by species name
            linewidth = 2) +
  # third layer: confidence interval
  geom_ribbon(data = preds, # data frame: -preds
              mapping = aes(
                        y = predicted,
                        ymin = conf.low,
                        ymax = conf.high,
                        fill = species_name), # colored by species name
               alpha = 0.1) + # made ribbon more transparent
  # created panel for each ant species
  facet_wrap(~ species_name,
             labeller = labeller(
             species_name = c(
                 "Crematogaster sjostedti" = "C. sjostedti",
                 "Crematogaster mimosae"   = "C. mimosae",
                 "Crematogaster nigriceps" = "C. nigriceps"))) +
  # changed to custom colors for lines and points
```

```
scale_color_manual(values = c(
  "Crematogaster sjostedti" = "darkblue",
  "Crematogaster mimosae" = "darkgreen",
  "Crematogaster nigriceps" = "brown4")) +
# changed to custom colors for ribbons
scale_fill_manual(values = c(
  "Crematogaster sjostedti" = "darkblue",
  "Crematogaster mimosae"   = "darkgreen",
  "Crematogaster nigriceps" = "brown4")) +
# set x axis range and breaks
scale_x_continuous(limits = c(0, 600),
                   breaks = c(0, 200, 400, 600)) +
# relabaled x- and y-axes
labs (x = "Tree height (cm)",
      y = "Probability of nest occurrence") +
# changed theme from default
theme_minimal() +
# removed legend
theme(legend.position = "none")
```



i. Write the caption for your figure

j. Calculate model predictions

```r
# what is the probability of nest occurence at a value of 600cm (the tallest tree in the data
ggpredict(model2,
          terms = "height_cm [600]")
```

```
# Predicted probabilities of case_control

height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.20 | 0.00, 0.99

Adjusted for:
* species_name = Crematogaster sjostedti
```

k. Interpret your results

Problem 3. Affective and exploratory visualizations

a. Comparing visualizations

I compared my final affective visualization (presented in Workshop 10; different from the original draft in Homework 3) to the exploratory visualizations from Homework 2:

- **Differences**: My affective visualization displays the logistic regression model (predicted probability of running as a function of sleep duration) within a hand drawn watercolor landscape, using illustrated runners and hourglasses as visual metaphors in place of geometric plot elements. The two exploratory visualizations from Homework 2 prioritized clarity and efficiency over emotional and creative aspects, using standard statistical geometry. The boxplot with jitter directly compares the distributions of workout duration across three workout types (categorical groups), while the scatterplot shows the relationship between two continuous variables (sleep duration and workout duration) with no model overlaid and no artistic embellishment.
- **Similarities**: All three visualizations use sleep duration or workout type on the x-axis and show some form of individual observations; the jittered open circles in the boxplot, the individual points in the scatterplot, and the binary data underlying the logistic curve in the affective visualization.

- **Patterns**: The scatterplot of sleep vs workout duration shows no clear trend; points are scattered without an obvious directional pattern, with most clustering around 60 minutes regardless of sleep duration. This makes sense because workout duration includes lifts and yoga, which the boxplot shows are consistently around 60-65 minutes regardless of other factors; meanwhile, runs have a much wider spread (roughly 36-75 minutes) and the lowest median of the three groups. Isolating runs as a binary outcome in the affective visualization therefore reveals a cleaner pattern (a steep positive relationship between sleep and probability of running, rising from near zero at 5 hours to near 1 at 10 hours) than using duration across all workout types.
- **Feedback**: NEED TO DO

b. Designing an analysis

I used a binomial GLM to ask whether sleep duration the previous night (continuous, measured in hours) predicts the probability that my workout was a run (binary, 1 = run and 0 = all other workout types). This analysis was appropriate because the response variable is binary, so a standardized linear model can not be used as the assumption of normally distributed errors is not met. Therefore, I used a GLM with a binomial error distribution because this test is specifically designed for response variables that have outcomes of 0/1. Sleep is a continuous predictor variable, so it works as the explanatory variable in my logistic regression.

c. Check any assumptions and run your analysis

Think about your data

The assumptions for a binomial GLM are met. First, each workout entry represents a separate day, so observations are independent. Second, there are binary outcomes, seen through the response variable run having 1 (a run occurred) and 0 (a run did not occur) outcomes. Finally, there is a linear relationship between transformed expected response in terms of link function and explanatory variables (WHAT DOES THIS MEAN – REPHRASE).

Write your model

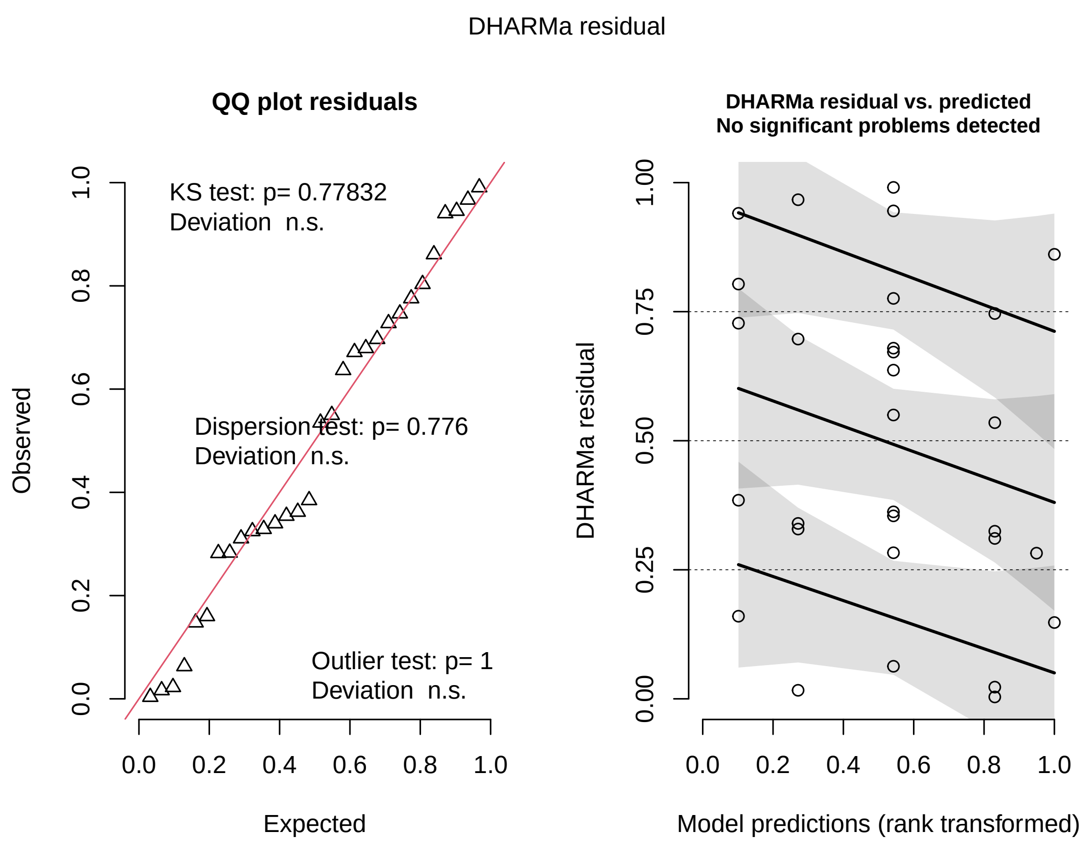
In words: Sleep duration the previous night predicts the occurrence of going on a run (yes = 1, no = 0).

In code:

```r
# fitted generalized linear model for running
# response: whether or not workout was a run (binary: 1/0)
# predictor: sleep the previous night (hours)
run_mod <- glm(
  # formula: binary run outcome ~ sleep hours
  is_run ~ sleep_hours,
  # data: my_data data frame
  data = my_data,
  # binomial error distribution for binary response
  family = binomial)
```

Look at the diagnostics

```r
# simulated residuals from GLM for diagnostic plots
plot(simulateResiduals(run_mod))
```



Information from left plot (QQ plot of residuals):

- **KS test**: Do my residuals follow a uniform distribution? → Generally yes
- **Dispersion test**: Are your residuals more dispersed than expected from a uniform distribution? → No
- **Outlier test**: Are there outliers in your data set that could be influencing model coefficients or predictions? → No

Information from right plot (residuals vs predicted values):

In visually assessing for randomly arranged residuals across the range of predicted values, DHARMa diagnostics flag issues in the 25th quartile. This is likely due to the small sample size, not because the model is wrong. Therefore, it it reasonable to say that residuals are randomly arranged across the range of predicted values.

Look at the model coefficients

```
# displayed model summary
summary(run_mod)
```

```

Call:
glm(formula = is_run ~ sleep_hours, family = binomial, data = my_data)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -8.7011     3.4964  -2.489   0.0128 *
sleep_hours   1.0873     0.4712   2.307   0.0210 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 36.652  on 29  degrees of freedom
Residual deviance: 28.046  on 28  degrees of freedom
AIC: 32.046

Number of Fisher Scoring iterations: 5
```

Note:

- all estimates are log odds, so calculate the probability of a run when sleep is 0 below using inverse logit

Characteristic	OR	95% CI	p-value
sleep_hours	2.97	1.37, 9.40	0.021

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Calculate an effect size

```
# calculated inverse logit
plogis(-8.7011)
```

```
[1] 0.000166375
```

Interpretation:

- inverse logit: 0.00016
- the probability of going on a run if sleep duration is 0 hours is 0.00016

```
# calculated odds ratio
gtsummary::tbl_regression(run_mod,
                          exponentiate = TRUE)
```

Interpretation:

- results table rendered at the top of this page
- odds ratio: 2.97
- 2.97 > 1 so the probability of going on a run increases by a factor of 2.97 with every 1 hour increase in sleep (95% CI: [1.37, 9.40])

Looking at the model predictions

- using probabilities to determine chance of getting certain outcome
- line represents the probability of 1 (going on a run) actually occurring

```
# what is the probability of going on a run after sleeping for 10 hours?
ggpredict(run_mod,
          terms = "sleep_hours [10]")
```

```
# Predicted probabilities of is_run

sleep_hours | Predicted |     95% CI
------------------------------------
         10 |      0.90 | 0.39, 0.99
```

```
# what is the probability of going on a run after sleeping for 4 hours?
ggpredict(run_mod,
          terms = "sleep_hours [4]")
```

```
# Predicted probabilities of is_run

sleep_hours | Predicted |     95% CI
------------------------------------
          4 |      0.01 | 0.00, 0.25
```

Note:

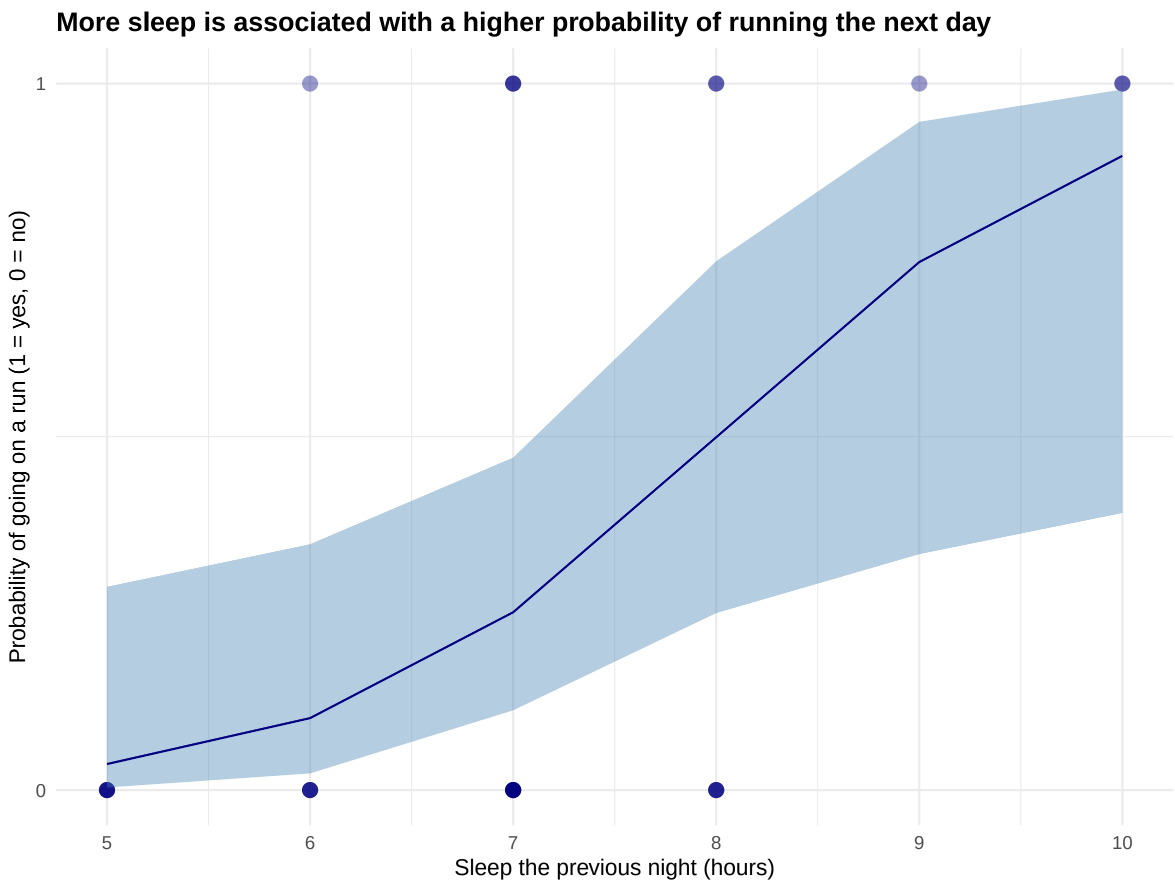
- If I slept for 10 hours, the probability of going on a run the next day is 0.90 (95% CI [0.39, 0.99]).
- If I slept for 4 hours, the probability of going on a run the next day is 0.01 (95% CI [0.00, 0.25]).

d. Create a visualization

Visualizing model predictions

```
# generated model predictions across range of sleep values in data set
mod_preds <- ggpredict(run_mod,
                       terms = "sleep_hours [5:10 by = 1]")

# base layer: ggplot with raw data
ggplot(my_data,             # data frame: my_data
       aes(x = sleep_hours, # x-axis: sleep duration
           y = is_run)) +   # y-axis: probability of running (0/1)
  # first layer: raw observations
  geom_point(size = 3,             # made points larger
             alpha = 0.4,          # made points more transparent
             color = "navyblue") + # changed points to navy blue
  # second layer: confidence interval ribbon from model predictions
```

```r
  geom_ribbon(data = mod_preds, # data frame: model predictions
              aes(x = x,
                  y = predicted,
                  ymin = conf.low,
                  ymax = conf.high),
              fill = "steelblue", # changed ribbon color to steel blue
              alpha = 0.4) + # made ribbon more transparent
  # third layer: logistic curve from model predictions
  geom_line(data = mod_preds,
            aes(x = x,
                y = predicted),
            color = "navyblue") +  # changed line color to navy
  # constrained y-axis to 0 and 1
  scale_y_continuous(limits = c(0, 1),
                     breaks = c(0, 1)) +
  # relabeled x- and y-axis
  # added title
  labs(title =
"More sleep is associated with a higher probability of running the next day",
       x     = "Sleep the previous night (hours)",
       y     = "Probability of going on a run (1 = yes, 0 = no)") +
  # changed theme from default
  theme_minimal() +
  # bolded title
  theme(plot.title = element_text(face = "bold"))
```



Describing the model

```
# displaying model in table
as_flextable(run_mod)
```

	Estimate	Standard Error	z value	Pr(>|z|)	
(Intercept)	-8.701	3.496	-2.489	0.0128	*
sleep_hours	1.087	0.471	2.307	0.0210	*

	Estimate	Standard Error	z value	Pr(>lzl)

*Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05*

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 36.65 on 29 degrees of freedom

Residual deviance: 28.05 on 28 degrees of freedom

e. Write a caption

Figure 1. Longer sleep duration is associated with a higher probability of going on a run the next day. Blue circles represent individual workout observations (0 = not a run, 1 = run). The navy blue line represents model predicted probability of running across the range of observed sleep values (5-10 hours), and the shaded blue ribbon represents the 95% confidence interval around model predictions. Data collected from personal fitness tracking records between 2026-04-14 and 2025-05-26.

f. Write about your results

I tended to go on runs more often on days following nights with more sleep. For every 1 hour increase in sleep the previous night, the odds of going on a run increased by a factor of 2.97 (95% CI: [1.37, 9.40], p = 0.021, $\alpha = 0.05$). After sleeping for 10 hours, the probability of going on a run the next day is 0.90 (95% CI [0.39, 0.99]), compared to a probability of 0.01 (95% CI [0.00, 0.25]) after sleeping for only 4 hours. This suggests that sleep duration has an influence on my likelihood of running.

g. Sharing your affective visualization

Presented visualization in Week 10 Workshop.